

Estimating spatial firm productivity spillovers using UK microdata

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DECLARATION

I, Aloysius Lip, hereby declare that the work presented in this dissertation is my own original work. Where information has been derived from other sources, I confirm that this has been clearly and fully identified and acknowledged. No part of this dissertation contains material previously submitted to the examiners of this or any other university, or any material previously submitted for any other assessment.

I declare the use of GenAI tools in an assistative role (UCL Category 2) for this submission. The model Gemini 3.10 Pro (DeepMind, 2026) was used for limited roles of: supporting the particular process of reviewing and refactoring code (data analysis), and as a support tutor (economic model and empirical approach).

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Date: 2nd September, 2026

CLASSIFICATION

This piece of research is primarily (please indicate as appropriate):

- ☒ an empirical/econometric study
- ☐ the examination of a theoretical problem
- ☐ a critical analysis of a policy issue
- ☐ an analytical survey of empirical and/or theoretical literature

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ABSTRACT

I estimate the productivity spillover occurring between colocated firms in the United Kingdom.

The productivity spillover, a spatial externality, underpins well-established economic phenomena around the development of urban clusters in the New Economic Geography literature, and is measured and heavily discussed in the evidence on the economic effects of urban density measured in the spatial econometrics literature, and underpinning regional growth policy. I apply econometric techniques that have been recently developed for valid identification when estimating spillover effects in a spatial group and network setting to firm Total Factor Productivity (TFP). This overcomes (Manski, 1993)’s ‘Reflection Problem’ and endogeneity of peer effects. It requires big data calculations of firm interactions, which I use modern data techniques to compute efficiently, preserving the advantages of Fame’s

high-observation balance sheet data for both its panel structure and its resulting high-significant estimates. Across my instrumented specifications run in static and dynamic specifications, I evaluate a spatial spillover of firm peers to any given target firm of an elasticity of (0.2), a significant proportion of GVA. These results imply that regional economic policy can have serious multiplier effects: more policy emphasis should be placed on raising the Total Factor Productivity of regional firms as its effects will spillover to its peers.

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1. INTRODUCTION

In section 2, I give an overview of the multiple intersecting fields from which this analysis draws. As background, I discuss how this analysis furthers contributions in New Economic Geography (NEG), urban econometric approaches, and how it may support emerging regional policy in the United Kingdom.

1.1. THE USEFULNESS OF ESTIMATING LOCAL PRODUCTIVITY SPILLOVERS

A productivity spillover is essential to the effects of New Economic Geography, however (Krugman, 1991) prefers not to assume this effect in their focus on the dynamic movement of firms. Instead, heterogenous cluster policies, such as initial external differences in the number of workers or firms are frequently assumed, with ‘forward and backward linkages’ driving colocation endogenously from there. This paper helpfully tests the presence of spillovers, providing another centrifugal force directly incentivising firms to colocate.

A revival of interest in regional, pro-second city policy has also taken place in the policies of the 2024 and 2026 governments in the United Kingdom. Regional economic policy declined strongly in the UK under the 1980s Thatcher government. . . This consensus was maintained throughout the 1990s-2000s New Labour period, where rapid economic growth known as ‘The Great Moderation’ provided no impetus for change. The consensus was broken in 2017, with the seeds of a regional-industrial strategy in 2017, emerging under the Conservative May government. Under the background of a UK-EU Referendum result (“Brexit”) which emphasised the sentiment of disaffection among ‘left-behind’ places across the UK which did not see the benefits of national aggregates of economic growth primarily concentrated in the UK’s capital, London. The national government launched ‘Local Industrial Strategies’ which sought to identify action to “boost productivity, earning power, and competitiveness” on a regional basis⁵. However, fewer of these strategies were fully developed and implemented by 2021. In 2026, the newly appointed Burnham government made the pledge for “good growth in every postcode”

1.2. ECONOMETRIC CONTRIBUTIONS

A key contribution by (Manski, 1993) is to argue that there is not just a direct, endogenous effect of the outcome of interest among peers, but also an exogenous effect of the characteristics of peers. However, Manski’s group model famously

but also a set of exogenous characteristics

It has been common to analyses of economic growth since (Marshall, 1920) to assume and qualitatively discuss some effect on firm-to-firm spillover of productivity. A productivity spillover underpins well-established economic phenomena such as the colocation of firms in urban clusters described in New Economic Geography literature such as (Krugman, 1991). It has been heavily analysed through industry and spatial density approaches in spatial and urban econometric literature such as (Glaeser et al., 1992).

2. LITERATURE

2.1. AGGLOMERATION LITERATURE

My investigation is rooted in Marshall (1920)’s taxonomy, which identifies three channels to productivity spillovers: (1) input-output costs, (2) labour pooling, and (3) knowledge spillovers.

⁵<https://researchbriefings.files.parliament.uk/documents/CBP-7682/CBP-7682.pdf>

There is no reason to think these are exhaustive. We might theorise that there may be competitive effects and cultural factors. Krugman also labels all these channels as 'centripetal' to an urban core; there are likewise 'centrifugal' factors that push firms away from the urban core, such as destructive (zero-sum) competition, congestion, and saturation costs. A further extension could develop models on these further channels and empirically test them. Furthermore, these productivity spillovers are incredibly salient to thinking about production structures.

In this investigation, I concentrate on (3) knowledge spillovers, tackling a difficult identification problem highlighted in Glaeser (2001). This is more relevant to the knowledge-based services economy that dominates the UK. Input-output costs and labour-pooling matter more for industrial manufacturing firms, especially the latter which was devised in response to the phenomenon emergence of industry towns.

Most agglomeration literature bundles all three channels together. Glaeser (2010) and Krugman (2010) identify how vital identifying knowledge spillovers are. That need to estimate them well only grows with the modern, knowledge economy of the UK. (Goldin, 2023) speculates that agglomeration effects have "increasing strength" attributable to the way in which "knowledge workers in particular become more productive when located near to one another. Information may be more available than ever, but the evidence suggests that it still tends to flow largely through in-person networks".

2.2. MODERN APPROACHES

Specific approaches to investigating knowledge spillovers via innovation, instrumenting via patent publications comes from US evidence, including (Jaffe, 1993), (Berkes and Ruben, 2021) and more recently (Bloom, Schankerman and Van Reenen, 2013).

Knowledge spillovers have been a well-evidence factor of within-firm managerial-derived productivity gains. A field-starting MIT study from Allen (1977) observed an exponentially declining curve depicting the frequency of communication between two individuals against their physical distance in an office.

2.3. ECONOMETRIC LITERATURE

- (Bramoullé et al., 2009)

3. DATA

3.1. FAME

I have used data from Fame for this investigation. Fame is a comprehensive set of financial datapoints from UK firms, with instruments appropriate for this analysis, in particular, number of employees and balance sheet data from which analysis-relevant aggregates can be derived. Fame is provided by Moody's Analytics, an international credit agency, and collects its data from UK firms via survey⁶.

Its international counterpart, Orbis, offers a similar service for international counterparts. This analysis could therefore be extended to offer insights comparing the spillover effects of UK firms to their international counterparts. For this investigation, the UK dataset is comprehensive and the most policy-relevant for this investigation.

Raw data was accessed from Fame in February 2025 by Professor Lars Nesheim for (Irastorza-Fadrique et al., 2026). and reused for this project. Raw data is provided on a series of a financial balance sheet indicators, including characteristics that are fixed for each firm, bucketed into *a1_ID* and *a5_misc* indicator tables, and annual data published on a calendar-year basis, bucketed into *a2_key_finance*, *a3_assets*, and *a4_profits* tables. I designed a series of Python scripts to ingest this data and store it in a 'DuckDB' database, separated into fixed and yearly datapoint tables. 'DuckDB' is an open-source, local, relational (SQL) architecture, designed for processing bulk queries across columns of variables, making it well-suited for econometric data analysis.

3.2. INGESTION AND CLEANING

(...)

All data is deflated to real terms, using 2024 as the base year. I used the GDP deflator for all balance sheet variables, except for *renumeration_employees* and its composites for which I used CPIH, as these are income-related aggregates with real spending power best captured by household purchasing baskets⁷.

All balance sheet data are in thousands ('000s) of GBP (£). As we estimate log relationships exclusively, this has no effect on parameter interpretation.

3.3. DERIVING PANEL AGGREGATES

Labour input is taken from the number of employees reported on the FAME balance sheet per firm. This dataset, like all others, does not include hours worked per employee on a firm-level that would offer a more granular productivity (GVA per hour worked) measures. An estimation in this setting would likely have to impute the measure from industry aggregates, which would likely confound the true firm-level signal.

Capital is Fame's *fixed_total* variable, which represents the sum of tangibles, intangibles, and other investments within the balance sheet, namely an extremely comprehensive measure of the firm's fixed capital. The average capital-output ratio is 6x within FAME when filtering for positive values Table (I), which is moderately higher than macroeconomic aggregates typically suggest likely due to accounting standard differences.

GVA is calculated as the sum of *ebitda* and *renumeration_employees*. *renumeration_employees* is the exact sum of *wages*, *social_security_costs*, and *pensions_costs* in the balance sheet. *ebitda*

⁶Analytics (2025)

⁷for National Statistics (2026)

TABLE I
Capital-output ratios in FAME corpus of firms

Variable	Description	Capital-output ratios
<i>tangibles</i>		3.53
<i>fixed_total</i>	<i>tangibles</i> + <i>intangibles</i> + <i>inv._other</i>	6.00
<i>total_assets</i>	<i>fixed_total</i> + <i>current_assets</i>	11.56

is in turn *turnover* minus intermediate inputs and is given in the Fame balance sheet for the vast majority of panel observations in our dataset. It is also possible to reconstruct *ebitda* via the sum of *profit_loss_pretax*, *interest_paid*, and *depreciation*, providing an alternative measure of GVA. We can expect the difference between these two to be accounting differences⁸ and additionally note that the alternate GVA measure also reduces panel observations from 1,085,520 to 618,588. For these reasons, I do not report TFP calculations with this second GVA measure which are regardless near-identical, though I resultingly refer to GVA throughout this paper as *gva_1*.

3.4. DERIVING 2-FACTOR TFP FOR METRIC SELECTION

To help guide the choice of capital and output (GVA) measure, I initially conduct the 2-factor estimation deriving the residual $z_{i,t}$ to decompose further in a second stage. This step is additionally helpful for validating the relationships about the derived aggregates from my data.

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + u_{i,t} \quad (1)$$

$$u_{i,t} = \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t}$$

$$z_{i,t} = \alpha_i + \gamma_t + \beta_0 + u_{i,t} \quad (2)$$

$$\hat{z}_{i,t} = y_{i,t} - \hat{\beta}_1 k_{i,t} - \hat{\beta}_2 l_{i,t} \quad (3)$$

I conduct the estimation using the log-transformed variables. I must include firm and time fixed effects and a constant if equation (6) is the true specification, otherwise the static panel OLS will inaccurately attribute firm and time variation to my classical factor inputs (explain more clearly). As outlined above, this analysis assumes factor input exogeneity.

The parameter coefficients give a labour income share exactly in line with ONS analysis of UK industry-level data over the same time period, particularly *corporate gross labour share* which is constructed from a similar sample to the coverage in Fame⁹.

Surprisingly, running the regression with *fixed_total* as the capital measure instead of *total_assets* almost entirely eliminates the capital share of income, or explanatory power of capital on variation in GVA. I leave it for future analysis why this dataset yields this phenomenon and for the rest of the analysis, I use *total_assets* as the primary capital measure.

As a demonstration, I include specifications which omit fixed effects. The capital share of income jumps on a specification with only time fixed effects, which is to be expected (explain).

⁸An OLS regression between the two, discarding 5% outliers, has a coefficient of 0.91 with a 0.000 p-value. This means *gva2* is slightly lower than *gva1*, with that impact mostly driven by top-end outliers.

⁹for National Statistics (2024)

TABLE II
Baseline 2-factor TFP estimation varying fixed effects

	(1.)	(2.)	(3.)	(4.)
	<i>gva</i> ₁	<i>gva</i> ₁	<i>gva</i> ₁	<i>gva</i> ₁
constant	2.995 (0.036)***	3.036 (0.031)***	1.970 (0.010)***	1.973 (0.010)***
$\ln(\text{employees})$	0.617 (0.005)***	0.614 (0.004)***	0.561 (0.002)***	0.561 (0.002)***
$\ln(\text{total_assets})$	0.286 (0.005)***	0.284 (0.004)***	0.421 (0.002)***	0.421 (0.002)***
<i>i</i> fixed effects	✓	✓		
<i>t</i> fixed effects	✓		✓	
Observations	1,045,164	1,045,164	1,045,164	1,045,164
R ² (excl. F.E.)	0.3313	0.3453	0.8094	0.8096
R ² (Overall)	0.7912	0.7900	0.8096	0.8096

All parameters reported with standard errors underneath. Significant estimates against an $H_0 : \beta = 0$ are indicated with $p < 0.1$ (*), $p < 0.05$ (**), and $p < 0.01$ (***).

TABLE III
Baseline 2-factor TFP estimation varying parameter measures

	(1.)	(2.)	(3.)	(4.)
	<i>gva</i> ₁	<i>gva</i> ₂	<i>gva</i> ₁	<i>gva</i> ₁
constant	2.995 (0.036)***	3.288 (0.056)***	4.751 (0.018)***	5.268 (0.070)***
$\ln(\text{employees})$	0.617 (0.005)***	0.611 (0.007)***	0.720 (0.004)***	0.628 (0.015)***
$\ln(\text{total_assets})$	0.286 (0.005)***	0.265 (0.007)***		
$\ln(\text{fixed_total})$			0.065 (0.002)***	
$\ln(\text{tangibles})$				0.090 (0.006)***
$\ln(\text{intangibles})$				0.025 (0.002)***
$\ln(\text{investments_other})$				0.007 (0.001)***
<i>i</i> fixed effects	✓	✓	✓	✓
<i>t</i> fixed effects	✓	✓	✓	✓
Observations	1,045,164	603,586	1,006,904	152,444
R ² (excl. F.E.)	0.3313	0.2764	0.2830	0.2699
R ² (Overall)	0.7912	0.7390	0.6751	0.7403

3.5. LOCATION GROUP CLASSIFICATION

The structure of the weighting matrix $\mathbf{W}$ is key to identification. W is an $n \times n$ matrix of weights where each column j denotes a weight w_{ij} applied on the variable of peer firm j , summing across rows i to reach a weighted aggregate regressor with an estimated effect on target firm i .

A first property of W is that its diagonal elements are all 0: a firm's TFP aggregate cannot be part of a weighted sum to an effect on its own TFP or GVA. This is common to peer effects literature such as (Moffitt and Caspi, 2001) and discussed in (Bramoullé et al., 2009).

Group classifiers impose an otherwise block diagonal structure on $\mathbf{W}$. The resulting $n \times n$ matrix has off-diagonal elements as the number of group members $n_g - 1$, which excludes the target firm i . The structure is given below¹⁰:

$$w_{ij} \in \mathbf{W}^{\text{group}} = \begin{cases} 0, & \text{if } i = j \\ \frac{1}{n_g - 1}, & \text{else if } (i, j) \in g \\ 0, & \text{otherwise} \end{cases}$$

$$\mathbf{\Gamma}_{s_1} \in \mathbf{W}^{\text{group}} = \{i, j \in g\}, s_1 : n_g$$

$$\mathbf{W}^{\text{group}} = \begin{pmatrix} \mathbf{\Gamma}_{s_1} & 0 & \dots \\ 0 & \mathbf{\Gamma}_{s_2} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}_{(n \times n)}$$

An example such matrix is $\mathbf{W}_1^{\text{group}}$, a (7×7) set of interactions between firms which have group classifiers of size $s_1 = 4$ and $s_2 = 3$ only.

$$\mathbf{W}_1^{\text{group}} = \begin{pmatrix} 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$$

As will be shown through discussions of spatial econometric properties, construction of a valid instrument requires calculation of $\mathbf{W}^2, \mathbf{W}^3, \dots, \mathbf{W}^k$.

TABLE IV
Summary statistics across location groups

Property	Number of groups	Unclassified	Unclassified (%)	Avg. peers per group
<i>ttwa</i>	228	567	0.4%	668.3
<i>pc4</i>	2,705	0	0.0%	56.3
<i>pc8</i>	49,963	0	0.0%	3.1
<i>lat_lon3</i>	48,540	507	0.3%	3.1
<i>lat_lon4</i>	53,193	507	0.3%	2.9
<i>lat_lon5</i>	53,246	507	0.3%	2.9

¹⁰Adapted from (Bramoullé et al., 2009)

3.6. LOCATION DISTANCE CALCULATIONS

Throughout this analysis, I use terminology of a productivity spillover occurring from a group or average of *peer* firms onto a *target* firm. This relationship is irreflexive. For groups it is transitive: if firms A and B share a spatial group, such as a travel-to-work area (*ttwa*), first-half of a UK postcode (*pc4*), or full postcode (*pc8*), then a firm C that is a peer of firm B must also be of firm A. For continuous distance measures, this is non-transitive. Firm B, a peer of firm A within its 5km radius, can clearly have its own peers that are within its own 5km radius that are not shared with A. Every firm is then its own overlapping group, or equivalently, a sparse, rather than block diagonal, firm-to-firm matrix¹¹.

(Cleaning address data).

Fame only reports each company's last-known registered and trading address, at the time of compilation of publication of its annual vintage (cite). As a result, final location data is only available at the firm level. We can interpret each firm as being static over the entire 20-year sample, with no movers. This seems very unlikely (expand). Instead, we must interpret the constructing address, and therefore firm-to-firm distance measures as a signal of firm pairs' true distance polluted by movers. This has 2 econometric consequences.

Firstly, firm fixed effects in all spatial specifications become extremely important. For all movers within the sample period, their firm location decision is endogenous to the subsequent productivity spillovers they experience and exert, for reasons discussed in Section 2 on New Economic Geography. (More on firm FEs) There is no solution to this for moves occurring out-of-sample, for instance, just before 2009.

Secondly, I acknowledge some attenuation to the productivity spillover estimate Peer firms who were close to the target firm during the firm-year observation who subsequently moved away within-sample will have an extreme distance recorded, and so be excluded from tagged address groups and truncated distance calculations. Peer firms who are recorded as close to the target firm as they subsequently moved to the location (More on direction of bias/positive selection?)

I first consider a model where each target has peers which share a spatial identifier, meaning they belong to the same travel-to-work-area (*ttwa*) using the 2011 definitions¹², the first half of a UK postcode *pc4*, and the full UK postcode *pc8*.

(Discuss and illustrate different forms of omega, with illustrative block diagonal matrices)

¹¹I do actually impose a block diagonal structure for reasons of computation as explained subsequently

¹²(Review ONS bulletin on TTWAs data source and describe clearly)

4. STATIC MODELS

I use a stylised 3-factor production model throughout this analysis. Y , A , K , and L are each firm-year observation's GVA, TFP, labour, and capital in levels. I use lower-case variables to indicate logged terms and bolded variables to indicate i -stacked vectors.

Log TFP, denoted by $z_{i,t}$, is the residual after the own-firm contribution of the two classical inputs. I decompose it to a constant across all observations, firm and time fixed effects and, most crucially for this investigation, the weighted sum of all peer log TFP values, $\sum_{j \neq i} w_{ij} z_{j,t}$. Different static weights, w_{ij} , are explored throughout this section. The objective is to estimate the structural coefficient δ_0 of spillover from peers to any given target firm.

$$Y_{i,t} = A_{i,t} \cdot K_{i,t}^{\beta_1} \cdot L_{i,t}^{\beta_2} \quad (4)$$

$$z_{i,t} = \ln(A_{i,t}) = y_{i,t} - \beta_1 k_{i,t} - \beta_2 l_{i,t} \quad (5)$$

$$= \alpha_i + \gamma_t + \beta_0 + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (6)$$

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (7)$$

$$= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \theta_0 \sum_{j \neq i} w_{i,j} y_{i,t} + \xi_0 \sum_{j \neq i} w_{ij} \cdot (\beta_1 k_{j,t} + \beta_2 l_{j,t}) + \epsilon_{i,t} \quad (8)$$

Equation (8) is a linear peer effects specification as given by (Manski, 1993). It is helpful to borrow the terminology from this literature think of $\begin{pmatrix} k_{i,t} & l_{i,t} \end{pmatrix} \in x_{i,t}$ as *exogenous* characteristics of firm i causally affecting its outcome $y_{i,t}$. There is an *endogenous* effect directly from peer outcomes $y_{j,t}$ with coefficient θ_0 . Peers also have an *exogenous* characteristics $x_{i,t}$, which have a direct impact on on the target's productivity $x_{i,t}$ via ξ_0 . For this section of models, I assume there are no *correlated* effects¹³. I use this peer effects terminology throughout this paper, noting to the reader the slight confusion this may cause with the more general but related concepts of endogeneity and exogeneity. The use of fixed effects has no effect on econometric interpretation of the peer effects model.

In a more general peer effects model, peer characteristics could independently affect the outcome of interest. We could imagine this as the choice level of capital or labour input in a peer j influencing the effectiveness, perhaps through management quality or hiring practices, of firm i . However, in this model I have strictly assumed all peer spillovers to occur through the weighted sum of $z_{j,t}$ as given in (6), so my true single parameter of interest is δ_0 . This is an exact linear combination of θ_0 and ξ_0 with no other confounding effects.

$$\delta_0 = \theta_0 - \xi_0 \quad (9)$$

While the classical production function (7) is limited to constant returns to scale, $\beta_1 + \beta_2 \leq 0$, a standard assumption across the growth and firm production literature, no such restrictions for the firm occur with the endogenous decomposition of TFP. A restriction on logged production (7) collectively on β_1 , β_2 , and δ_0 is difficult to justify. As with other external effects, there is no reason to believe that market factor prices and the firm decision-rule for spatial allocation in view of these spillover effects, $w_{ij} z_{j,t}$, is socially optimal. A peer firm j with extremely high and growing TFP has little interest in siting themselves in the centre of a cluster that are struggling with TFP growth, yet the external impacts of that firm would be significant.

¹³ (how would that enter the specification?)

4.1. STATIC PANEL IDENTIFICATION

Consistency requires the plausible exogeneity of regressors used in structural equation (8) from the error term $\varepsilon_{i,t}$.

Capital and labour are assumed to be exogenous in this static model and exhibit unit factor substitutability. In this section, we can broadly interpret this as meaning factor input decisions are predetermined.

$$x_{i,t} = \begin{pmatrix} k_{i,t} & l_{i,t} \end{pmatrix} \quad K_{i,t} = I_{i,t} + dK_{i,t-1} \quad (10)$$

$$I_{i,t} = \mathbb{E}[I_{i,t}^* | \Omega_{t-1}] \quad (11)$$

$$L_{i,t} = \mathbb{E}[L_{i,t}^* | \Omega_{t-1}] \quad (12)$$

$$\begin{aligned} \mathbb{E}[\varepsilon_{i,t} | \Omega_{t-1}] &= 0 \\ \implies \mathbb{E}[\varepsilon_{i,t} | x_{i,t}] &= 0 \end{aligned} \quad (13)$$

The peer effects regressors are endogenous, making valid estimation with the peer effects regressors is more complicated. To show this requires stacking w_{ij} target-peer weights to an $n \times n$ interactions matrix, $\mathbf{W}$. I invert the *endogenous* component to give a $(\mathbb{I} - \theta_0 \mathbf{W})^{-1}$ coefficient on all variables.

$$\theta_0 < 1 \implies |\mathbb{I} - \theta_0 \mathbf{W}| \neq 0 \quad (14)$$

The inversion is permissible because I can expect a restriction to apply, ruling out iteratively increasing shocks¹⁴. This is directly comparable to the lag polynomial and MA inversion procedure in autoregressive time series analysis¹⁵.

$$\mathbf{y}_t = \boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + \theta_0 \mathbf{W} \mathbf{y}_t + \xi_0 \mathbf{W} \cdot (\beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t) + \boldsymbol{\varepsilon}_t \quad (15)$$

$$\begin{aligned} &= (\mathbb{I} - \theta_0 \mathbf{W})^{-1} (\boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + \xi_0 \mathbf{W} \cdot (\beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t) + \boldsymbol{\varepsilon}_t) \\ &= \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \cdot (\boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + \xi_0 \mathbf{W} \cdot (\beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t) + \boldsymbol{\varepsilon}_t) \end{aligned} \quad (16)$$

$$= \frac{\gamma_t + \beta_0}{1 - \theta_0} \boldsymbol{\iota} + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \boldsymbol{\alpha} + \beta \mathbf{x}_t + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{x}_t \beta \cdot (\theta_0 + \xi_0) + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \boldsymbol{\varepsilon}_t \quad (17)$$

$$y_{j,t} = \cdots + \underbrace{\sum_{k=0}^{\infty} [\theta_0^k \mathbf{W}^{k+1}]_{(j, \cdot)} \begin{pmatrix} \mathbf{k}_t & \mathbf{l}_t \end{pmatrix} \beta \cdot (\theta_0 + \xi_0)}_{\text{exogenous}_j} + \underbrace{\sum_{k=0}^{\infty} [\theta_0^k \mathbf{W}^k]_{(j, \cdot)} \cdot \boldsymbol{\varepsilon}_t}_{\text{error transformation}_j} \quad (18)$$

It is the primary econometric challenge of this analysis to overcome this spatial endogeneity. The reduced-form equation (18) confirms that within the *endogenous* effect $\theta_0 \sum_{j \neq i} w_{ij} y_{j,t}$, the peer GVA component $y_{j,t}$ is indeed endogenous, containing a linear combination of all other firms' $\boldsymbol{\varepsilon}_t$, implying:

$$\mathbb{E}[y_{j,t} \varepsilon_{i,t}] \neq 0 \quad (19)$$

The ‘Reflection Problem’ poses a second issue as outlined in (Manski, 1993). Equation

¹⁴ $\theta_0 > 1$ would imply an exogenous TFP shock to firm k would increase TFP in its peer firm j more than 1x, which would increase TFP in that firm's peer i more than 1x, and so on until the effect multiplies to infinity

¹⁵Equation (17) stacks the $x_{i,t}$ *exogenous* parameter definition given above and vectorises the coefficients $\beta = (\beta_1 \quad \beta_2)'$. A full algebraic derivation from lines (16) to (17) is available in

(18) shows that $\mathbf{W}\mathbf{y}_{t,i}$ is an exact linear combination of all other regressors and variation in $y_{j,t}$ is indistinguishable in reduced-form from variation in all other regressors. This hinders identification of θ_0 , and in turn δ_0 .

There are two solutions. I could either proceed with an extremely strong exogeneity assumption, (20), which is most likely satisfied by total orthogonality between all contemporaneous idiosyncratic errors hitting peer firms (21), in turn making the regressor $y_{j,t}$ exogenous allowing direct estimation of (7). Model (1.) takes this assumption and estimates the second-stage (6). For identification, this implies that any productivity shock $\epsilon_{j,t}$ occurring in peer firm j , if it is contemporaneously correlated with TFP $z_{i,t}$ in target firm i must be attributable to the peer effect coefficient δ_0 via θ_0 . This precludes any contemporaneous, correlated independent TFP shocks, which might occur on both the global and local level. We should assess these results with the heavy caveat that the exogeneity assumptions required are unlikely to be credible.

$$\mathbb{E}[\varepsilon_i | \theta_0^k \mathbf{W}^k \begin{pmatrix} \alpha & \gamma_t \boldsymbol{\iota} & \beta_0 \boldsymbol{\iota} & \mathbf{k}_t & \mathbf{l}_t \end{pmatrix}] = 0, \forall k$$

$$\mathbb{E}[\varepsilon_{i,t} | \mathbf{W}^k \boldsymbol{\varepsilon}_t] = 0, \forall i, j \quad (20)$$

$$\Longleftarrow \mathbb{E}[\epsilon_{i,t} \epsilon_{j,t}] = 0, \forall k \quad (21)$$

Instead, as recommended in (Bramoullé et al., 2009), the most defensible approach is to instrument the *endogenous* variable, $y_{j,t}$ through the *exogenous* parameters $\mathbf{W}^2 \mathbf{x}_t, \mathbf{W}^3 \mathbf{x}_t \dots \mathbf{W}^k \mathbf{x}_t \forall k > 2$. These are be valid instruments under certain testable properties of $\mathbf{W}$. This is intuitive: we can see through equation (18) that a linear combination of all firms' contemporaneous capital and labour inputs, $\mathbf{W}^{k+1} \begin{pmatrix} \mathbf{k}_t & \mathbf{l}_t \end{pmatrix} \forall k > 1$ is an exogenous instrument. It has relevance, meaning $\mathbb{E}[y_{j,t} \cdot [\mathbf{W}^k]_{(j, \cdot)} \mathbf{x}_t] \neq 0, \forall k > 2$ as long as $\theta_0 \neq 0$ and $\beta_0 + \xi_0 = \delta_0 \neq 0$. These two conditions are unsurprising and highly intuitive - note that they differ slightly from the more complex setup given in (Bramoullé et al., 2009), this is because I derive the *exogenous* effect in (8) as the accounting composition of TFP and so β is necessarily included as a post-multiplication factor for the *exogenous* effect allowing factorisation, which is absent from the general model.

The testable conditions are:¹⁶

We can also note that $\mathbf{W}\mathbf{x}_t$ is also a valid instrument but unhelpful in this estimation as it is also a regressor as the direct *exogenous* effect.

A summary of the identification rules listed in (Bramoullé et al., 2009) is as follows:

I take this approach in Models (2, 3, 4). The instruments to employ are as follows: (...)

4.2. LINEAR-IN-MEANS GROUPED PEERS

Assuming exogeneity assumptions (20) and (21) hold, the following reduced-form equation can be estimated with a simple panel OLS.

$$z_{i,t} = \alpha_i + \gamma_t + \beta_0 + \delta_0 \sum_j w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (\text{from 6})$$

The initial static model applies simple time-invariant weights w_{ij} following the interactions structure of $\mathbf{W}^{\text{group}}$. As has been indicated by the sum of peers $j \neq i$, I adopt a 'leave-one-out' approach in line with the zero-diagonal structure of $\mathbf{W}^{\text{group}}$ described. Intuitively, this is the unweighted average of peer TFP within a spatial ring. This weighting matrix is the classic set of

¹⁶Bramoullé et al. (2009)

TABLE V
Spatial econometric identification rules for weighting matrix W

Structure	Required condition	Identified	Instruments
<i>Group structure</i>			
Proposition 2	$\beta\theta_0 + \xi_0 \neq 0$		
Linear-in-Means (LMM) ^(a)	Target is own peer	$\times$ $W^2 = W$	—
Leave-one-out ^(b)	Target excluded from peer mean	$\times$	—
	Same group size	$W^2 = \frac{1}{s-1}\mathbb{I} + \frac{s-2}{s-1}W$	
Varying group sizes ^(c)	Target excluded from peer mean ≥ 2 group sizes	$\checkmark$	$s_g - 1$
<i>Network structure</i>			
Proposition 1	$\mathbb{I}, W, W^2$ linearly independent $\beta\theta_0 + \xi_0 \neq 0$	$\checkmark$	W^2x
Intransitive triad	$\exists i, j, k : W_{ij} > 0, W_{jk} > 0, W_{ik} = 0$	$\checkmark$	W^2x
Transitive networks	$W^2 \neq 0$	$\checkmark$	W^2x
Proposition 3	$W^2 \neq 0; \beta\theta_0 + \xi_0 \neq 0$	$\checkmark$	W^2x
$W^2 = 0$	$\alpha \neq 0$	$\checkmark$	Wx
<i>Adding correlated effects</i>			
Proposition 4	$\mathbb{I}, W, W^2, W^3$ linearly independent $\beta\theta_0 + \xi_0 \neq 0$	$\checkmark$	$(\mathbb{I} - W)W^2x$
Corollary 1: group structure	≥ 3 group sizes	$\checkmark$	$(\mathbb{I} - H)W^2x$
Complete bipartite / star	$W^3 = W$	$\times$	—
Proposition 5	$W^3 = \lambda_0\mathbb{I} + \lambda_1W + \lambda_2W^2$	$\checkmark$	$(\mathbb{I} - H)W^2x$
	$\text{rank}(\mathbb{I} - W) < n - 1$		
	$2\lambda_0 + \lambda_1 + 1 \neq 0$		
Full rank spatial polynomial	$\text{rank}(\mathbb{I} - W) = n - 1$	$\times$	None

Adapted from Bramoullé et al., 2009. (a) Manski, 1993, (b) Moffitt and Caspi, 2001, (c) fei Lee, 2007. **(Increase row size here; review varying group sizes, $\alpha \neq 0$)**

weights applied for the 'Linear-in-Means (LMM)' Model common to the peer effects literature.

I discussed group assignment in section 3.6. Firms have been labelled using their location features to symmetric, transitive, non-overlapping groups by *ttwa* (largest) to *lat_lon5* (smallest), with the latter meaning two firm entries shared a latitude-longitude identifier to 5 decimal places. Reviewing the summary statistics, I opted for full postcodes, *pc8* as the group identifier for Table VI.1.. This was mostly driven by full data availability, and the consistency of postcode definitions with the larger group *pc4*. Deriving a subgroup by latitude-longitude match within the postcode seemed fruitless, as in many cases it would identify the exact same firms, a move from the average number of peers from 3.1 to 2.9, and was a similar number of groups, with non-cohesive definitions.

The results below show model (1), where peers and targets share a postcode.

TABLE VI
Static panel linear-in-means using spatial-group TFP averages

	(1.) <i>gva</i> ₁	(2.) <i>gva</i> ₁	(3.) <i>tfp</i>	(4.) <i>tfp</i>
constant	2.568 (0.038)***	2.573 (0.058)***	2.322 (0.012)***	2.289 (0.014)***
$\ln(\text{total_assets})$	0.238 (0.005)***	0.238 (0.005)***		
$\ln(\text{employees})$	0.651 (0.005)***	0.651 (0.005)***		
$\ln(\text{gva1_pc8})$	0.213 (0.004)***	0.212 (0.004)***		
$\ln(\text{total_assets_pc8})$	-0.049 (0.002)***	-0.049 (0.002)***		
$\ln(\text{employees_pc8})$	-0.124 (0.004)***	-0.123 (0.004)***		
$\ln(\text{gva1_pc4_d})$		0.010 (0.004)***		
$\ln(\text{total_assets_pc4_d})$		-0.005 (0.002)**		
$\ln(\text{employees_pc4_d})$		-0.001 (0.004)		
$\ln(\text{gva1_ttwa_d})$		-0.007 (0.004)**		
$\ln(\text{total_assets_ttwa_d})$		-0.000 (0.003)		
$\ln(\text{employees_ttwa_d})$		0.009 (0.007)		
<i>tfp_pc8</i>			0.246 (0.004)***	0.245 (0.004)***
<i>tfp_pc4_d</i>				0.005 (0.001)***
<i>tfp_ttwa_d</i>				0.004 (0.002)***
<i>i</i> fixed effects	✓	✓	✓	✓
<i>t</i> fixed effects	✓	✓	✓	✓
Observations	732,689	724,978	732,689	725,040
R ² (excl. F.E.)	0.3994	0.3987	0.0297	0.0300
R ² (Overall)	0.8161	0.8159	0.1071	0.1063

As an extension, I regress the multiple, non-overlapping spatial groups at once. Multiple spatial autoregressors do not change the reflection problem: the moving average inversion's coefficient merely becomes the weighted sum of 3 independent matrices $\delta_g \mathbf{W}_g$, for a generic set of weights $\mathbf{W}_g$.

$$z_{i,t} = \alpha_i + \gamma_t + \beta_0 + \sum_{g=1}^3 \delta_g \sum_{j \neq i} w_{ij,g} z_{j,t} + \varepsilon_{it} \quad (22)$$

$$\mathbf{z}_t = \sum_{k=0}^{\infty} (\delta_1 \mathbf{W}_1 + \delta_2 \mathbf{W}_2 + \delta_3 \mathbf{W}_3)^k \cdot (\boldsymbol{\alpha} + \gamma_t \mathbf{1} + \beta_0 \mathbf{1} + \boldsymbol{\varepsilon}_t) \quad (23)$$

The weighting rules for these multiple, non-overlapping groups is to zero (...) These matrices are orthogonal, justifying the result found above that the coefficient on δ_0 is identical in the 1-regressor model δ_0 to the 3-regressor model for δ_1 .

$$w_{ij} \in \mathbf{W}_g = \begin{cases} 0, & \text{if } i = j \\ 0, & \text{if } w_{ij,k} \neq 0, \exists k < g \\ \frac{1}{n_g - 1}, & \text{else if } (i, j) \in g \\ 0, & \text{otherwise} \end{cases} \quad (24)$$

$$\sim \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & 0 & 0 & 0 \\ \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & 0 & 0 & 0 \\ \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & 0 & 0 & 0 \end{pmatrix}$$

The parameter results suggest a strong correlation between the TFP of a target firm and its peers. Peers firms in the local postcode have an elasticity of 0.29, significant at the 1% level, on the target firm's TFP. However, since I failed to show exogeneity of $z_{j,t}$ within the regressor, we cannot describe this as causal. Consider omitted variable bias from a local demand shock (...)

4.3. STATIC DISTANCE DECAY

To improve on the LMM specification, I make a modification where every peer firm is weighted by its influence on the target by the reciprocal of the continuous distance metric calculated in 3.6 $w_{ijt} = f(d)$. This is stylistic of a declining peer influence further away, representing a (lower frequency of social collisions, barriers created via transport and time costs, and an increased difficulty of observation of management practices).

$$z_{it} = \alpha_i + \gamma_t + \beta_0 + \beta_1 \sum_{j \neq i} w_{ijt} z_{jt} + \varepsilon_{it} \quad (25)$$

$$\mathbf{z}_t = (\alpha_i + \gamma_t + \beta_0) \mathbf{1} + \beta_1 \sum_{j \neq i} \Omega_t \mathbf{z}_t + \varepsilon_t \quad (26)$$

$$w_{ijt} \in \Omega_t = \begin{cases} \frac{1}{d_{ij}}, & (1) \text{ gravity} \\ \frac{1}{d_{ij}^2} & (2) \text{ N.E.G.} \\ \exp -d_{ij} \cdot k & (3) \text{ exponential decline} \end{cases} \quad (27)$$

This represents a likely more accurate declining effect of productivity spillover, rather than the special-case group mean where every firm in an enclosed area, for example, the target's postcode, has an equal influence on the target's TFP. In addition, it allows for better identification following Bramoullé et al. (2009). (explain Bramouille2009, patterns of) Ω_t In essence, the LMM specification only contains 3 non-overlapping groups from its location tags. The distance-decayed specification contains as many non-overlapping groups as firms.

TABLE VII
Static panel using continuous distance-decay weighted
TFP average of peers

	(1) <i>tfp</i>	(2) <i>tfp</i>	(3) <i>tfp</i>
constant	2.479 (0.029)***	2.586 (0.029)***	2.417 (0.032)***
$\ln(\textit{employees})$	-0.009 (0.004)**	-0.009 (0.004)**	-0.009 (0.004)**
$\ln(\textit{nb_peers})$	0.013 (0.004)***	0.012 (0.004)***	0.016 (0.004)***
<i>tfp_wav</i> ₁	0.168 (0.003)***		
<i>tfp_wav</i> ₂		0.135 (0.003)***	
<i>tfp_wav</i> ₃			0.184 (0.005)***
<i>i</i> fixed effects	✓	✓	✓
<i>t</i> fixed effects	✓	✓	✓
Observations	1,067,013	1,067,013	1,067,013
R ²	0.0101	0.0095	0.0040

This distance-decay specification revises down the estimate from peers to a an elasticity of 0.168, lower than the results given by the LMM specification. This is likely due to (...) In addition, I introduce controls (...)

5. DYNAMIC MODELS

I have discussed how the static model requires a strong strict exogeneity assumption. I showed in subsection 4.1 that cross-sectional orthogonality is a **(necessary condition)** to meet this assumption with a peer-effects regressor.

$$\begin{aligned}
y_{i,t} &= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \\
z_{i,t} &= \alpha_i + \gamma_t + \beta_0 + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \\
\mathbb{E}[\varepsilon_{i,t} \varepsilon_{j,t}] &= 0, \quad \forall t \\
\implies \mathbb{E}[\varepsilon_{i,t} | x_{i,t}] &= 0, \quad \forall t \quad x_{i,t} = \begin{pmatrix} k_{i,t} & l_{i,t} & \sum_{j \neq i} w_{ij,t} z_{j,t} \end{pmatrix} \\
\implies \mathbb{E}[\Delta \varepsilon_{i,t} | \Delta x_{i,t}] &= 0, \quad \forall t
\end{aligned}$$

In this section, I introduce a dynamic panel specification to relax these assumptions, permitting weak cross-sectional correlation between peer errors and endogeneity of the contemporaneous regressors. Crucially, this specification can still be identified even in its panel reduced form, which requires first differences to eliminate fixed effects, by the use of internal instruments. This follows econometric theory developed by (Anderson and Hsiao, 1981) and estimator techniques developed in (Arellano and Bond, 1991) and furthered by (Blundell and Bond, 1998) which improve efficiency.

There are good reasons to think this weak cross-sectional correlation exists. A local demand shock **(...)**

5.1. INSTRUMENTAL IDENTIFICATION UNDER NO ERROR AUTOCORRELATION

The estimator given in (Arellano and Bond, 1991) takes the regression equation as given above and allows for endogeneity. Internal instruments of all the endogenous regressors are found, meaning their lagged first differences. Identification is improved in the System GMM approach given in (Blundell and Bond, 1998) by also including the first lagged, not differenced, lag of the endogenous regressors which meets the usual instrument conditions, as an instrument. It is this System GMM approach that I follow here, so I find the first lagged, not differenced, lag of regressor in x_{it} and its corresponding set of first difference lagged instruments, which in practice just start from $\Delta x_{i,\tau-j}$ for any j if $x_{i,\tau}$ is a valid instrument.

I commence finding these valid instruments by assuming the structural specification (7) exhibits no autocorrelation in its error term $\varepsilon_{i,t}$, though I now allow for moderate cross-sectional correlation across firms.

$$\mathbb{E}[\varepsilon_{i,t} \varepsilon_{i,t-1}] = 0, \quad \forall t, t-1 \quad (28)$$

$$\mathbb{E}[x_{i,t} \varepsilon_{i,t}] \neq 0, \quad \exists i \quad (29)$$

$$\mathbb{E}[\varepsilon_{i,t} \varepsilon_{j,t}] \neq 0, \quad \exists i, j \quad (30)$$

$$\mathbb{E}[\varepsilon_t] = 0, \mathbb{E}[\varepsilon_t^2] = \sigma_t^2, \mathbb{E}[\varepsilon_t | \Omega_{t-1}] = 0, \quad \forall i \quad (\text{w.n.}) \quad (31)$$

$$\begin{aligned}
x_{i,t} &= \begin{pmatrix} k_{i,t} & l_{i,t} & \sum_{j \neq i} w_{ij,t} z_{j,t} \end{pmatrix} \\
\Delta x_{i,t} &= \begin{pmatrix} \Delta k_{i,t} & \Delta l_{i,t} & \Delta \sum_{j \neq i} w_{ij,t} z_{j,t} \end{pmatrix}
\end{aligned}$$

Taking each regressor in turn, I relax the exogeneity assumption of the standard factors in the reduced-form equation, $\Delta k_{i,t}$ and $\Delta l_{i,t}$. I assume a new optimal firm decision on capital

based on information, denoted by Ω_t which includes $\varepsilon_{i,t}$ for its respective t . This approach allows the discovery of valid instruments which are exogenous to $\varepsilon_{i,t}$ and $\Delta\varepsilon_{i,t}$. I justify an exclusion restriction for $\Delta k_{i,t-j} \forall j > 1$, as it is predetermined, and $\Delta l_{i,t-k}, k > 2$, as it is flexible.

$$\Delta y_{i,t} = \Delta\gamma_t + \beta_1 \Delta k_{i,t} + \beta_2 \Delta l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} \Delta z_{j,t} + \Delta\varepsilon_{i,t} \quad (32)$$

$$K_{i,t} = I_{i,t} + dK_{i,t-1} \quad (33)$$

$$I_{i,t} = \mathbb{E}[I_{i,t}^* \mid \Omega_{t-1}] \quad (34)$$

$$L_{i,t} = \mathbb{E}[L_{i,t}^* \mid \Omega_t] \quad (35)$$

$$\varepsilon_{i,t} \in \Omega_t$$

$$\implies K_{i,t} \propto \epsilon_{t-1}$$

$$\implies L_{i,t} \propto \epsilon_t$$

$$\Delta\varepsilon_{i,t} = \varepsilon_{i,t} - \varepsilon_{i,t-1}$$

$$\Delta k_{i,t} = k_{i,t} - k_{i,t-1} \propto \varepsilon_{i,t-1} \varepsilon_{i,t-2}$$

$$\implies \mathbb{E}[\Delta k_{i,t} \Delta \epsilon_t] \neq 0$$

$$\implies \mathbb{E}[\Delta k_{i,t-1} \Delta \epsilon_t] = 0, \mathbb{E}[k_{i,t-1} \varepsilon_t] = 0 \quad (36)$$

$$\Delta l_{i,t} = l_{i,t} - l_{i,t-1} \propto \varepsilon_{i,t}, \varepsilon_{i,t-1}$$

$$\implies \mathbb{E}[\Delta l_{i,t-2} \Delta \epsilon_t] = 0, \mathbb{E}[l_{i,t-2} \varepsilon_t] = 0 \quad (37)$$

Devising an internal instrument for $\Delta \sum_{j \neq i} w_{ij,t} z_{j,t}$ requires a similar evaluation of the spatial econometric properties of $z_{j,t}$ as in the static problem. z_{it} is a linear combination of $I_{i,t}$ and $L_{i,t}$. Importantly, the deep lag strategy instrumenting via $z_{i,t-2}$ is valid even with this section's new cross-sectional correlation in equation 30.

$$\mathbf{z}_{i,t} = \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^k \cdot (\boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \boldsymbol{\varepsilon}_t) \quad (38)$$

$$\Delta \mathbf{z}_t = \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^k \cdot (\Delta \gamma_t \boldsymbol{\iota} + \Delta \boldsymbol{\varepsilon}_t) \quad (39)$$

5.2. AUTOREGRESSIVE SPECIFICATION

A dynamic specification grants the flexibility to add past realisations of variables to the decomposition of TFP, $z_{i,t}$. This increases the number of regressors in reduced form, improving the plausibility of its exogeneity with $\varepsilon_{i,t}$.

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \phi_1 y_{i,t-1} + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (40)$$

The above equation is in structural form in the 1-stage regression, but I can find an equivalent in the 2-stage regression with $z_{i,t}$ as the dependent variable.

$$z_{i,t} = \alpha_i + \gamma_t + \beta_0 + \phi_1 y_{i,t-1} + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (41)$$

$$= \alpha_i + \gamma_t + \beta_0 + \phi_1 \cdot (\beta_1 k_{i,t-1} + \beta_2 l_{i,t-1} + z_{i,t-1}) + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (42)$$

For system GMM, I need to show that the reduced-form first difference regressors are exogenous.

$$\Delta y_{i,t} = \Delta \gamma_t + \beta_1 \Delta k_{i,t} + \beta_2 \Delta l_{i,t} + \sum_{k=1}^p \phi_k \Delta y_{i,t-k} + \delta_0 \sum_{j \neq i} w_{ij,t} \Delta z_{j,t} + \Delta \varepsilon_{i,t} \quad (43)$$

$$\Delta z_{i,t} = \Delta \gamma_t + \phi_1 \cdot (\beta_1 \Delta k_{i,t-1} + \beta_2 \Delta l_{i,t-1} + \Delta z_{i,t-1}) + \delta_0 \sum_{j \neq i} w_{ij} \Delta z_{j,t} + \Delta \varepsilon_{i,t} \quad (44)$$

The first-order autoregressor, $\Delta \ln(y_{it})$ is endogenous.

$$\begin{aligned} \Delta \varepsilon_{i,t} &= \varepsilon_{i,t} - \varepsilon_{i,t-1} \\ \Delta y_{i,t-1} &= \dots + \Delta \varepsilon_{i,t-1} \\ &= \dots + \varepsilon_{i,t-1} - \varepsilon_{i,t-2} \\ &\implies \mathbb{E}[\Delta y_{i,t-1} \Delta \varepsilon_{i,t}] \neq 0 \\ \Delta y_{i,t-2} &= \dots + \varepsilon_{i,t-2} - \varepsilon_{i,t-3} \\ &\implies \mathbb{E}[\Delta y_{i,t-1} \Delta \varepsilon_{i,t}] = 0 \end{aligned} \quad (45)$$

This is confirmed by the Anderson-Rubin (AR(1)) test in my results, which rejects the null hypothesis at the 1% level for all models, suggesting that first-order lags of log GVA are a weak instrument (Why? Shouldn't they fail exclusion restriction, not instrument correlation). In addition, I assume that capital, $k_{i,t}$ is pre-determined, as investment is chosen one period ahead, in line with standard Perpetual Inventory Method approaches¹⁷ (Review predetermined material. Additionally, explain the logic of instrumenting for capital)

To overcome this endogeneity, I introduce deeper autoregressive lags as instrumental variables, as proposed by (Anderson and Hsiao, 1981). I use 3 autoregressive instrument lags starting from $t-2$: $\ln(y_{i,t-2}), \ln(y_{i,t-3}), \ln(y_{i,t-p-3})$ for log GVA, and 2 lags for capital $\ln(k_{i,t-p-1}), \ln(k_{i,t-p-2})$. An Anderson-Rubin (AR(2)) test shows that deep lags are a valid instrument for log GVA. The test applied to the second lag of $\ln(gva_1)$ fails to reject the null hypothesis of for any specification, so I assume it is a valid instrument.

Additionally, using multiple deep lags allows for overidentification: in this basic AR(1) specification, I have $k = 4$ regressors and $l = 7$ instruments. I could increase this. Fame is rich on the time dimension, with up to 19 lags at most for a firm, and a median of 6 lags, so I could safely provide more lags of regressors as instruments. Further lags is likely to have a decreasing marginal effect on explanatory power and additionally ((...))

(Discuss introducing a new weighting method using past firm similarity.)

I estimate the equation fulfilling the vector of moment conditions, (46) via GMM.

$$\mathbb{E} \left[\Delta \varepsilon_{it} \cdot \begin{pmatrix} \Delta \ln(y_{i,t-2}) \\ \Delta \ln(y_{i,t-3}) \\ \Delta \ln(y_{i,t-4}) \\ \Delta \ln(k_{i,t-2}) \\ \Delta \ln(k_{i,t-3}) \\ \Delta \sum_j w_{ijt} z_{jt} \end{pmatrix} \right] = 0 \quad (46)$$

As my regressors are over-identified, I use 2-step GMM to improve efficiency. (I should introduce a control (basically a diff-in-diff?) of peers from the same industry across the country (outside

¹⁷Noting that this is ordinarily done on a quarterly basis, however, due to data limitations we adopt the same thing here even though Fame provides observations annually.

the spatial bubble).)

TABLE VIII
Dynamic panel: all instrumented variables

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$
constant	0.653 (0.345)	0.031 (0.215)	-0.656 (0.970)
$\ln(gva_1)_{t-1}$	0.006 (0.001)***	0.003 (0.001)***	0.004 (0.002)
tfp_wav_{1t-1}			-0.017 (0.012)
$\ln(employees)$	0.523 (0.051)***	0.655 (0.027)***	0.599 (0.075)***
$\ln(total_assets)$	0.310 (0.030)***	0.298 (0.030)***	0.283 (0.036)***
tfp_wav_1	0.832 (0.087)***	0.887 (0.081)***	1.251 (0.303)***
Hansen Test	✓ (0.049)**	✓ (0.015)**	X (0.941)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.119)	X (0.879)	X (0.117)
Instruments	35	35	52
i fixed effects	✓	✓	✓
t fixed effects	✓	✓	✓
Observations	713,339	713,339	713,339

Model includes a constant as the baseline for t fixed effects.

Finally, a Sargan-Hansen J-test is implemented to check if these instruments are over-identified, meaning the moment conditions cannot be fulfilled with 95% confidence. In all models, the J-test fails to reject the null hypothesis, meaning the model's moment conditions are jointly correct.

5.3. INSTRUMENTAL IDENTIFICATION UNDER ERROR AUTOCORRELATION

I appear to have strong identification given assumptions (28) and (30). The instruments above exhibit strong correlation under Anderson-Rubin tests and exclusion restriction has been derived. However, there exists a niche case of diagonally-correlated errors, namely despite (28) and (30), I ought to consider the following:

$$\mathbb{E}[\varepsilon_{i,t}\varepsilon_{j,t-k}] \neq 0, \quad \exists i, j, k \quad (47)$$

This is not as implausible as it may appear due to this analysis' peer effects setting. There may be some local omitted shock ν_t , that impacts peer firm A's TFP via ε_t and impacts target firm B's TFP after a lag via ε_{t-k} . This could be a price shock with random adjustment (or)

Additionally, relaxing (28) to allow for weak error autocorrelation between limited $\varepsilon_{i,t}$ and $\varepsilon_{i,t-k}$ for limited k proves a similar problem.

(Firstly, I test for error autocorrelation.)

TABLE IX
Dynamic panel using instrumented continuous
distance-decay weighted TFP

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$
constant	1.371 (0.134)***	1.577 (0.144)***	1.064 (0.103)***
$\ln(gva_1)_{t-1}$	0.005 (0.001)***	0.005 (0.001)***	0.005 (0.001)***
$\ln(employees)$	0.696 (0.038)***	0.689 (0.037)***	0.709 (0.040)***
$\ln(total_assets)$	0.259 (0.041)***	0.266 (0.041)***	0.247 (0.044)***
tfp_wav_1	0.503 (0.028)***		
tfp_wav_2		0.423 (0.023)***	
tfp_wav_3			0.618 (0.042)***
Hansen Test	X (0.417)	X (0.459)	X (0.246)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.746)	X (0.600)	X (0.945)
i fixed effects	✓	✓	✓
t fixed effects	✓	✓	✓
Observations	713,339	713,339	713,339

Model includes a constant as the baseline for t fixed effects.

5.4. PANEL VECTOR AUTOREGRESSION

Another mechanism offered by this dynamic setting is to extremely flexibly introduce additional lags of my peer effect term, $\sum_{j \neq i} w_{i,j,t} z_{j,t}$.

$$y_{i,t} = \beta_0 + \alpha_i + \gamma_t + \sum_{k=1}^p \phi_k y_{i,t-k} + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \sum_{k=1}^q \delta_k \sum_{j \neq i} w_{i,j,t} z_{j,t} + \varepsilon_{i,t} \quad (48)$$

The model now has the form of a panel VAR(p,q) with internal instruments, where p refers to the autoregressive lags on the dependent variable $y_{i,t}$ and q refers to the regressor lags on $\Delta \sum_{j \neq i} w_{i,j,t} z_{j,t}$ chosen by my specification.

Following above reasoning, lags of $\Delta \sum_{j \neq i} w_{i,j,t} z_{j,t}$ of a higher order than 2 are exogenous to my reduced-form specification. However,

5.5. ALTERNATIVE WEIGHTS ON PEER TFP

$$y_{i,t} = \beta_0 + \alpha_i + \gamma_t + \sum_{k=1}^p \phi_k y_{i,t-k} + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \sum_{k=1}^q \delta_k \sum_{j \neq i} w_{i,j,t} z_{j,t} + \varepsilon_{i,t} \quad (49)$$

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APPENDIX 1: SOLVING THE REFLECTION PROBLEM

$$\begin{aligned}
& \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k (\mathbf{x}_t \boldsymbol{\beta} + \mathbf{W} \mathbf{x}_t \boldsymbol{\beta} \cdot \xi_0) \\
&= \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \mathbf{x}_t \boldsymbol{\beta} + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{x}_t \boldsymbol{\beta} \cdot \xi_0 \\
&= \mathbf{x}_t \boldsymbol{\beta} + \sum_{k=1}^{\infty} \theta_0^k \mathbf{W}^k \mathbf{x}_t \boldsymbol{\beta} + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{x}_t \boldsymbol{\beta} \cdot \xi_0 \\
&= \mathbf{x}_t \boldsymbol{\beta} + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{x}_t \boldsymbol{\beta} \cdot \theta_0 + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{x}_t \boldsymbol{\beta} \cdot \xi_0 \\
&= \mathbf{x}_t \boldsymbol{\beta} + \left(\sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{x}_t \right) \boldsymbol{\beta} \cdot (\theta_0 + \xi_0) \\
\mathbf{y}_t &= \frac{\alpha}{1 - \theta_0} \boldsymbol{\iota} + \mathbf{x}_t \boldsymbol{\beta} + \left(\sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^{k+1} \mathbf{x}_t \right) \boldsymbol{\beta} \cdot (\theta_0 + \xi_0) + \sum_{k=0}^{\infty} \theta_0^k \mathbf{W}^k \boldsymbol{\varepsilon}_t
\end{aligned}$$

APPENDIX 2

TABLE X
Dynamic panel: higher-order autoregressive lags

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$	(4) $\ln(gva_1)$
constant	1.371 (0.134)***	1.446 (0.843)	0.208 (2.578)	0.459 (2.844)
$\ln(gva_1)_{t-1}$	0.005 (0.001)***	0.019 (0.104)	0.128 (0.295)	-0.020 (0.256)
$\ln(gva_1)_{t-2}$		0.002 (0.003)	0.004 (0.101)	0.094 (0.300)
$\ln(gva_1)_{t-3}$			-0.001 (0.006)	0.036 (0.162)
$\ln(gva_1)_{t-4}$				-0.003 (0.004)
$\ln(employees)$	0.696 (0.038)***	0.762 (0.097)***	0.736 (0.149)***	0.659 (0.323)*
$\ln(total_assets)$	0.259 (0.041)***	0.190 (0.090)*	0.248 (0.121)*	0.291 (0.305)
tfp_wav_1	0.503 (0.028)***	0.548 (0.059)***	0.508 (0.076)***	0.466 (0.184)*
Hansen Test	X (0.417)	X (0.922)	X (0.881)	X (0.826)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	X (0.140)	X (0.212)
AR(2) Test	X (0.746)	X (0.925)	X (0.802)	X (0.668)
i fixed effects	✓	✓	✓	✓
t fixed effects	✓	✓	✓	✓
Observations	713,339	603,058	511,396	433,185

TABLE XI
Dynamic panel: higher-order regressor lags 1-5

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$	(4) $\ln(gva_1)$	(5) $\ln(gva_1)$
constant	1.124 (0.113)***	0.956 (0.101)***	0.743 (0.093)***	0.584 (0.091)***	0.543 (0.094)***
$\ln(gva_1)_{t-1}$	0.002 (0.001)*	0.003 (0.001)***	0.005 (0.001)***	0.005 (0.001)***	0.005 (0.001)***
tfp_wav_{1t-1}	0.066 (0.006)***	0.077 (0.007)***	0.069 (0.007)***	0.062 (0.007)***	0.061 (0.008)***
tfp_wav_{1t-2}		0.052 (0.005)***	0.055 (0.006)***	0.050 (0.007)***	0.051 (0.007)***
tfp_wav_{1t-3}			0.043 (0.005)***	0.048 (0.006)***	0.049 (0.006)***
tfp_wav_{1t-4}				0.030 (0.005)***	0.038 (0.006)***
tfp_wav_{1t-5}					0.028 (0.005)***
$\ln(employees)$	0.686 (0.036)***	0.695 (0.038)***	0.670 (0.040)***	0.644 (0.044)***	0.665 (0.052)***
$\ln(total_assets)$	0.270 (0.040)***	0.260 (0.042)***	0.287 (0.045)***	0.317 (0.049)***	0.294 (0.057)***
tfp_wav_1	0.508 (0.028)***	0.514 (0.029)***	0.492 (0.030)***	0.469 (0.033)***	0.480 (0.038)***
Hansen Test	X (0.492)	X (0.968)	X (0.453)	X (0.084)*	X (0.080)*
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.279)	X (0.612)	X (0.801)	X (0.854)	X (0.729)
i fixed effects	✓	✓	✓	✓	✓
t fixed effects	✓	✓	✓	✓	✓
Observations	713,339	603,058	511,396	433,185	365,908

TABLE XII
Dynamic panel: higher-order regressor lags 6-10

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$	(4) $\ln(gva_1)$	(5) $\ln(gva_1)$
constant	0.421 (0.082)***	0.388 (0.081)***	0.270 (0.077)***	0.199 (0.083)*	0.116 (0.093)
$\ln(gva_1)_{t-1}$	0.004 (0.001)***	0.004 (0.001)**	0.003 (0.001)*	0.002 (0.002)	0.002 (0.002)
tfp_wav_{1t-1}	0.055 (0.008)***	0.056 (0.008)***	0.047 (0.008)***	0.042 (0.009)***	0.036 (0.011)**
tfp_wav_{1t-2}	0.045 (0.007)***	0.045 (0.007)***	0.040 (0.007)***	0.037 (0.007)***	0.035 (0.009)***
tfp_wav_{1t-3}	0.046 (0.006)***	0.046 (0.006)***	0.039 (0.006)***	0.033 (0.007)***	0.029 (0.008)***
tfp_wav_{1t-4}	0.034 (0.006)***	0.031 (0.006)***	0.024 (0.006)***	0.019 (0.007)**	0.017 (0.008)*
tfp_wav_{1t-5}	0.029 (0.005)***	0.030 (0.006)***	0.023 (0.006)***	0.019 (0.006)**	0.017 (0.006)**
tfp_wav_{1t-6}	0.018 (0.004)***	0.024 (0.005)***	0.021 (0.005)***	0.019 (0.005)***	0.017 (0.006)**
tfp_wav_{1t-7}		0.019 (0.005)***	0.017 (0.005)***	0.015 (0.006)**	0.011 (0.007)
tfp_wav_{1t-8}			0.013 (0.004)**	0.016 (0.005)**	0.017 (0.006)**
tfp_wav_{1t-9}				0.010 (0.005)*	0.013 (0.005)*
$\ln(employees)$	0.628 (0.049)***	0.636 (0.053)***	0.582 (0.055)***	0.548 (0.061)***	0.523 (0.073)***
tfp_wav_{1t-10}					0.016 (0.006)**
$\ln(total_assets)$	0.334 (0.054)***	0.325 (0.058)***	0.384 (0.061)***	0.421 (0.067)***	0.448 (0.080)***
tfp_wav_1	0.452 (0.036)***	0.457 (0.038)***	0.420 (0.040)***	0.397 (0.044)***	0.380 (0.052)***
Hansen Test	X (0.097)*	X (0.122)	X (0.450)	X (0.637)	X (0.435)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.924)	X (0.962)	X (0.568)	X (0.507)	X (0.450)
i fixed effects	✓	✓	✓	✓	✓
t fixed effects	✓	✓	✓	✓	✓
Observations	307,394	256,016	210,798	171,125	136,527

TABLE XIII
Dynamic panel: higher-order VAR(2,4) structures

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$	(4) $\ln(gva_1)$	(5) $\ln(gva_1)$	(6) $\ln(gva_1)$
constant	1.164 (0.573)*	0.498 (1.561)	0.398 (1.667)	1.494 (0.428)***	0.333 (1.469)	1.351 (0.307)***
$\ln(gva_1)_{t-1}$	-0.011 (0.117)	0.081 (0.280)	-0.028 (0.251)	-0.129 (0.081)	0.087 (0.276)	-0.125 (0.063)*
$\ln(gva_1)_{t-2}$	-0.000 (0.003)	0.006 (0.111)	0.076 (0.385)	0.004 (0.002)	-0.001 (0.111)	0.006 (0.002)**
$\ln(gva_1)_{t-3}$		-0.001 (0.005)	0.034 (0.156)		-0.002 (0.004)	
$\ln(gva_1)_{t-4}$			-0.002 (0.004)			
tfp_wav_1	0.551 (0.052)***	0.504 (0.078)***	0.475 (0.201)*	0.525 (0.049)***	0.488 (0.073)***	0.524 (0.050)***
tfp_wav_{1t-1}	0.099 (0.102)	0.005 (0.255)	0.082 (0.217)	0.196 (0.073)**	-0.005 (0.249)	0.189 (0.055)***
tfp_wav_{1t-2}	0.060 (0.011)***	0.036 (0.099)	-0.019 (0.342)	0.072 (0.012)***	0.046 (0.102)	0.070 (0.010)***
tfp_wav_{1t-3}				0.056 (0.009)***	0.039 (0.020)	0.065 (0.008)***
tfp_wav_{1t-4}						0.042 (0.006)***
$\ln(employees)$	0.737 (0.099)***	0.720 (0.146)***	0.643 (0.382)	0.650 (0.075)***	0.705 (0.148)***	0.657 (0.079)***
$\ln(total_assets)$	0.209 (0.088)*	0.253 (0.119)*	0.307 (0.367)	0.272 (0.074)***	0.271 (0.119)*	0.266 (0.079)***
Hansen Test	X (0.971)	X (0.855)	X (0.849)	X (0.999)	X (0.851)	X (0.232)
AR(1) Test	✓ (0.002)***	X (0.124)	X (0.163)	✓ (0.000)***	X (0.111)	✓ (0.000)***
AR(2) Test	X (0.889)	X (0.878)	X (0.775)	X (0.092)*	X (0.837)	✓ (0.030)**
i fixed effects	✓	✓	✓	✓	✓	✓
t fixed effects	✓	✓	✓	✓	✓	✓
Observations	603,058	511,396	433,185	511,396	511,396	433,185

APPENDIX 3